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Total-order Sobol' shares at the joint knee. The spread is 84-99% epistemic – model ignorance, not randomness – and the two terms that govern the bondline are both engineering judgment.

84-99%

of the variance in peak flux and bondline temperature, across all four designs, is epistemic.

It is removed by evidence, not
by design margin.

bondline, at the knee

0.638

TPS conductivity – a placeholder property (no material is specified)

- T3 judgment

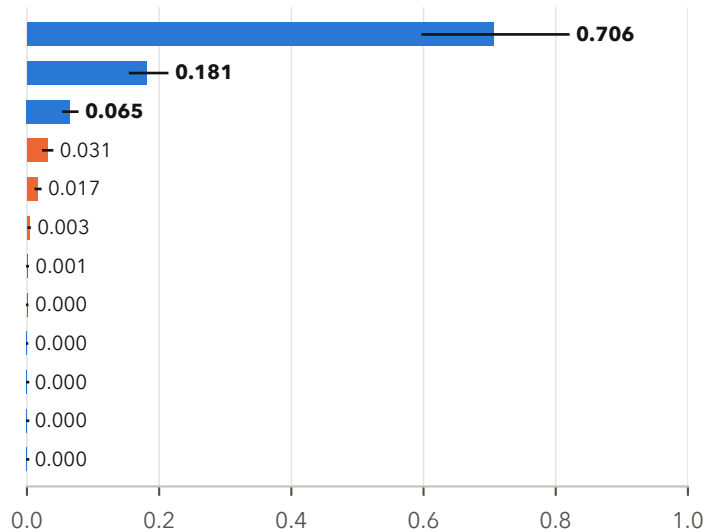
0.262

heating above 86 km – a continuum
correlation in transitional flow,
 $\times 0.5-1.5 \cdot T_3$ judgment

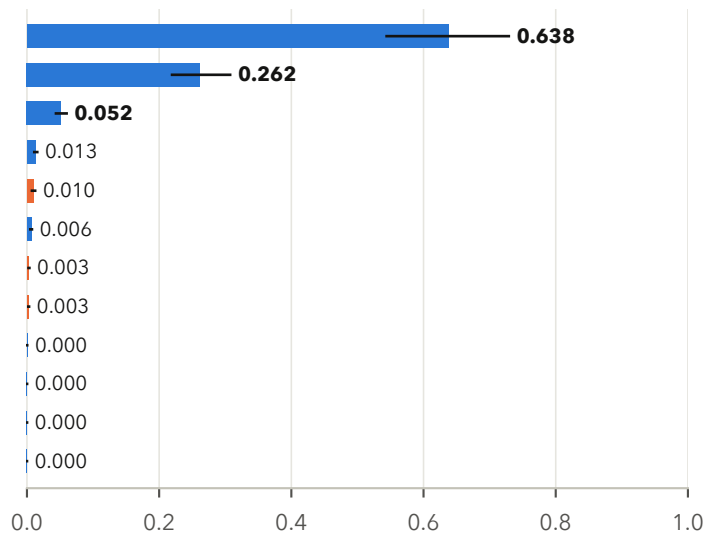
Peak flux at the knee: 0.706 of its variance is the disagreement between two NASA primaries on the effective nose radius (T1, resolvable) – and 0.000 on the hemispherical baseline, where the two agree.

- T1 effective nose-radius model (Ellison vs Zoby & Sullivan)
- T1 Sutton-Graves constant $\pm 4\%$
- T3 drag: perfect-gas model form $\pm 5\%$
- T3 vehicle mass $\pm 2\%$ (1σ)
- T3 atmospheric density dispersion
- T3 delivered entry angle
- T1 drag: CFD discretisation
- T2 drag: assumed base band
- T1 drag: GP surrogate error
- T3 TPS specific heat $\pm 10\%$
- T3 TPS conductivity $\pm 15\%$
- T3 heating above 86 km, $\times 0.5$ -1.5

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total-order Sobol' index S_T [-], bar = point estimate, whisker = 95% CI



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how to read the bars

- epistemic – a fact nobody has pinned down; evidence removes it

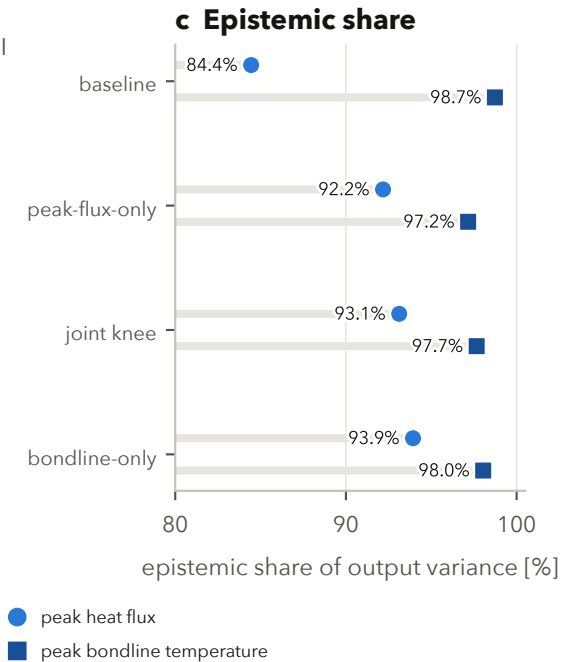
■ aleatory – real variability;
only margin absorbs it

■ T1 primary source

T2 secondary source

☐ T3 engineering judgment

tier = how the input's distribution
was sourced (configs/uncertainty.yaml);
7 of 12 inputs are judgment



- peak heat flux
- peak bondline temperature